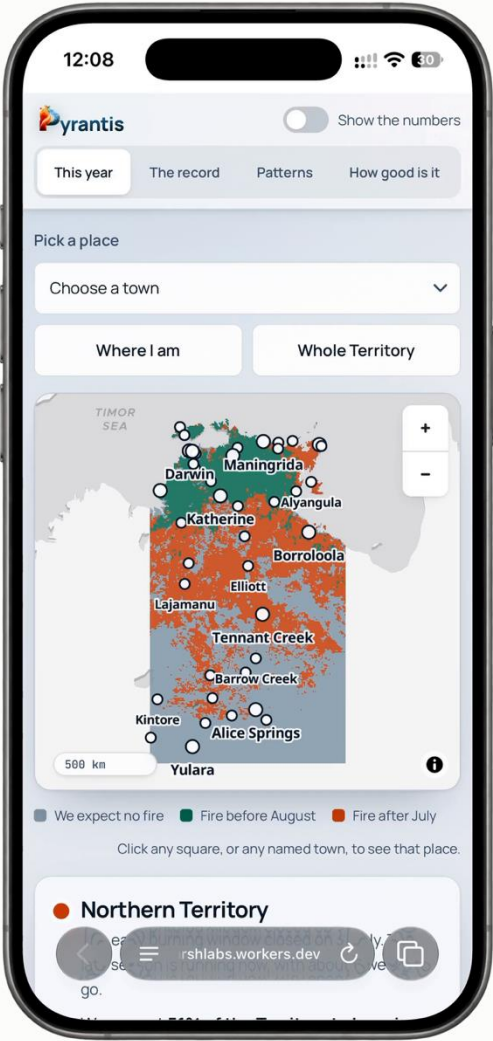


PYRANTIS

Which country burns in the Northern Territory, and when

Multi-class classification of 46,445 map squares across 26 fire years, and a forecast for 2026.

UNIT	Machine Learning, Artificial Intelligence and Algorithms	TEAM	
UNIT CODE	PRT565	Harsh Rastogi	386401
ASSESSMENT	Assessment 3 — Presentation	Saira Zafar	407193
GROUP	Group 85	Tharushi Wimalachandra	387594



Most of the Territory burns. The question is when.

Across 2000 to 2025, roughly a third of the Northern Territory carried a fire in any given year. Nearly a million square-years of record say the same thing: burning is not the exception here, it is the season.

1,207,570

square-years of fire record

46,445

map squares, five kilometres across

EVERY SQUARE, EVERY YEAR, 2005 TO 2025



■ no fire ■ fire before August ■ fire after July

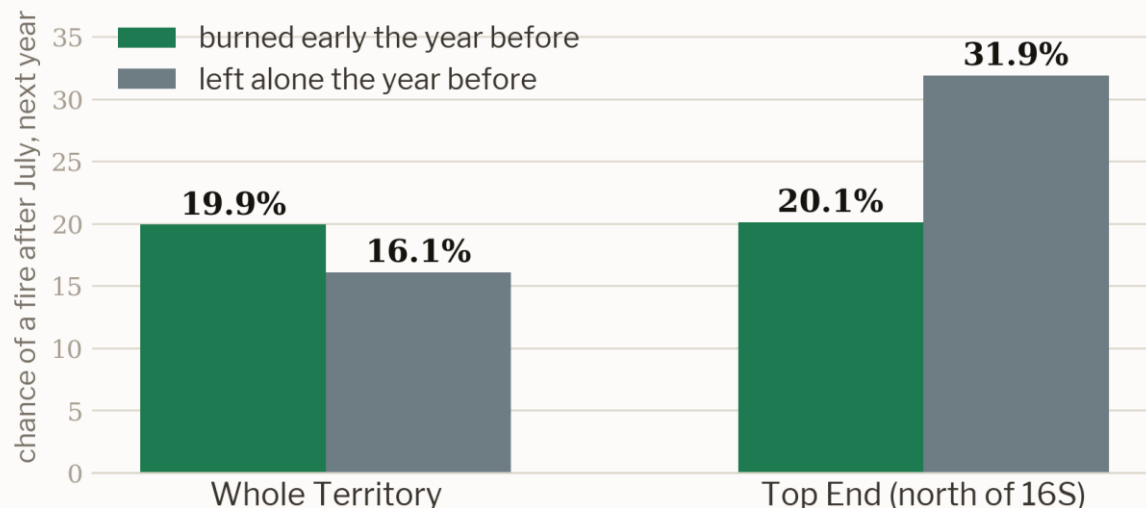
A fire in June and a fire in October are different events.

Before August

The country is still damp from the wet. Fires are cool, they creep, and they leave a patchwork. Rangers light these deliberately to break up the fuel.

After July

The grass has cured. Fires run with the wind, burn hot enough to kill canopy, and take out fences, stock and country in a single afternoon.

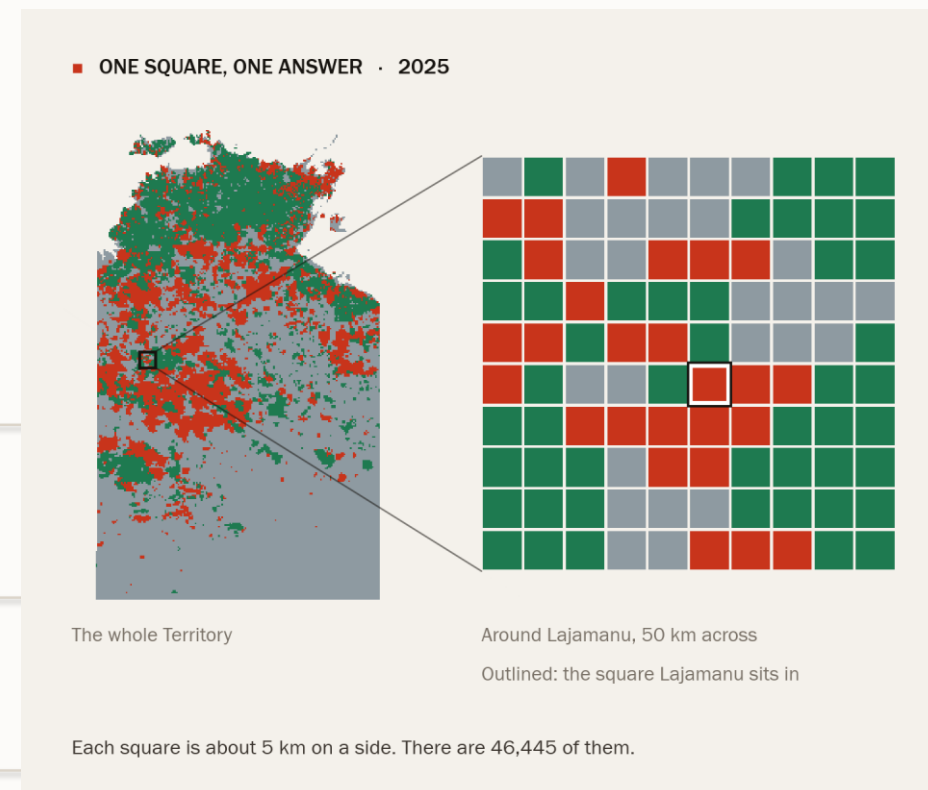


And it compounds.

In the Top End, country burned early is far less likely to carry a late fire the next year — 20.1% against 31.9% for country left alone. Territory-wide the comparison reverses, because the arid south burns on a different clock. One number for the whole map would have hidden both facts.

For every square, for every year: which of three?

- unburnt no fire detected that year
- early first detected on or before 31 July
- late first detected after 31 July



The 31 July line is the one the Territory's own fire management uses, so the classes mean something to the people who would act on them.

Three public sources, joined on one grid.

NAFI

North Australian Fire Information

250 m burnt-area rasters, 2000 to 2025. Each pixel carries the month it was detected — which is what makes an early/late label possible at all.

SILO

Queensland Government Data Drill

Daily rainfall, temperature, humidity and radiation, interpolated from Bureau of Meteorology stations to the same 0.05° grid.

MODIS

MOD13Q1 via Google Earth Engine

Sixteen-day vegetation index. How green the country got over the wet season is how much fuel is standing when the dry arrives.

All three are open data, used under their own licences and cited in the report. Nothing in this project sits behind a login.

0.05°

grid cell, about 5 km

2000–2025

years of fire record

50

features per square-year

966,819

rows used for modelling

02

ACT TWO

How we built it

Turning satellite scars into labels, deciding what the model is allowed to know, and proving it never saw the answer.

Saira Zafar

From burnt pixels to one answer per square.

1 Read the scar

Each NAFL pixel holds the month it burned, 1 to 12, or zero for no fire.

2 Pool to a square

Every 20×20 block of pixels becomes one 0.05° cell, and we count what fraction burned early and what fraction burned late.

3 Decide the class

A square counts as burnt only if at least 5% of it burned, and is only used at all if at least half its pixels were readable. Whichever season holds more of the burnt area names the year.

The check that mattered

Labels are the one thing no metric can catch. So we tested them against fire history the Territory already knows: 2011, the year after the 2010–11 La Niña, comes out as the heaviest on record at 65.7% of the map burnt. 2020 comes out the lightest at 15.9%. Both match the published record. If the labels had been wrong, every number after this would have been wrong too.

Only what a forecaster could actually know on 30 April.

30 April

the cut-off, every feature, every year

The forecast has to be useful before the fire season starts, so every one of the 50 features is built from information available by the end of April in the year being predicted — and nothing later.

Fire history

26 features

Years since the last fire and the last late fire, how often it burnt over 3, 5, 10, 15, 20 and 25 years, what it did last year, and what its neighbours did.

Weather

14 features

Wet-season rainfall, heat, humidity deficit and sunlight, plus the dry season before it, and how far the wet sat from its own local normal.

Greenness

7 features

How green the country got over the wet, where it peaked, how fast it greened up, and how far April sat above or below normal.

Place

3 features

Longitude, latitude and elevation, so the model can tell the Top End from the arid south without being handed the regions.

Two rules, because a leak would flatter every number after it.

A feature that quietly reads the year it is meant to predict will score beautifully and mean nothing. We did not trust ourselves to avoid that by being careful, so we tested for it twice.

RULE ONE

Recomputation

Every fire-history feature is rebuilt from years strictly before the target year, in one function that both the training path and the forecast path call. A second implementation would have drifted, and the drift would have been silent — the forecast has no label to check itself against.

RULE TWO

Scrambling

We shuffled one year's labels and rebuilt the features. If anything were reading the answer, the scores would have collapsed or held suspiciously steady. They moved exactly as much as chance says they should.

Both checks are scripts in the repository, not claims in a slide.

Split by time, never at random.

Fire years are not independent draws. A random shuffle would put 2024 in training and 2023 in test, and the model would be reading its own future. So the record is cut in chronological order and the last three years are never touched until the end.

Train

2005 – 2019

fifteen years

Tune

2020 – 2022

three years

Test

2023 – 2025

three years, seen once

2005

2025

And once was once. The test years were scored a single time, at the end, after every decision about features and models had already been made. Tuning against them would have turned the held-out years into another training set.

Six models, two baselines, and one correction for imbalance.

Decision tree	the readable one
Random forest	many trees, voting
Naive Bayes	the probabilistic floor
Multilayer ANN	a dense network
LSTM	reads the 26-year sequence
Gradient boosting	trees that correct each other

Against two baselines: “whatever happened last year” and “always say unburnt”. A model that cannot beat those has learned nothing worth having.

Why class weighting

Late fires are the ones that matter and the rarest thing on the map. Left unweighted, every model quietly stopped predicting them — it is cheaper to be right about the common answer. Weighting the classes makes that trade expensive.

GRADIENT BOOSTING, F1 ON LATE FIRES

0.420
unweighted

0.553
weighted

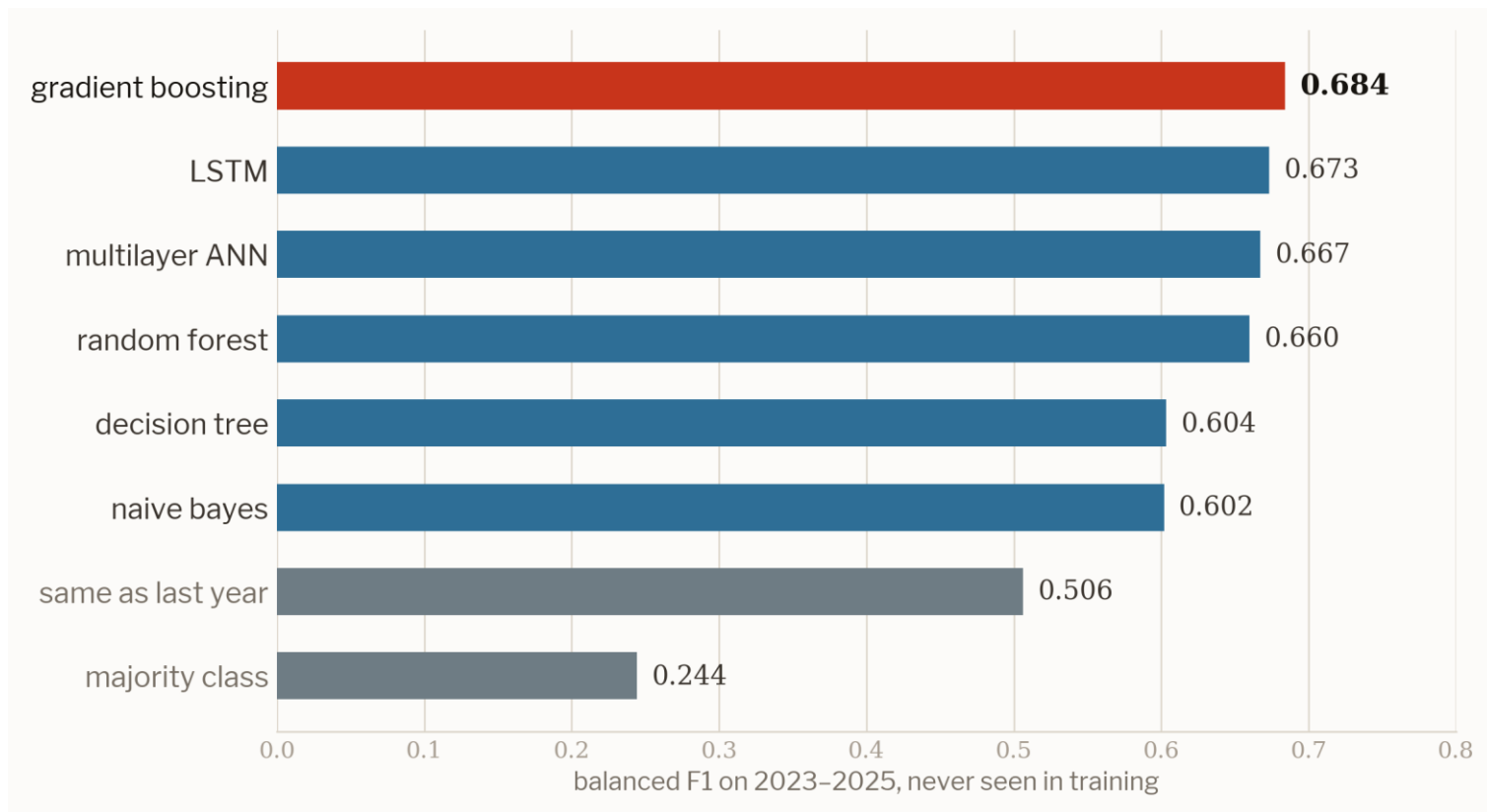
ACT THREE

Does it actually work?

Scores on years the models never saw, sixty-six separate re-tests, and the twenty-three experiments behind the one we kept.

Tharushi Wimalachandra

Gradient boosting wins, and every model beats the baselines.



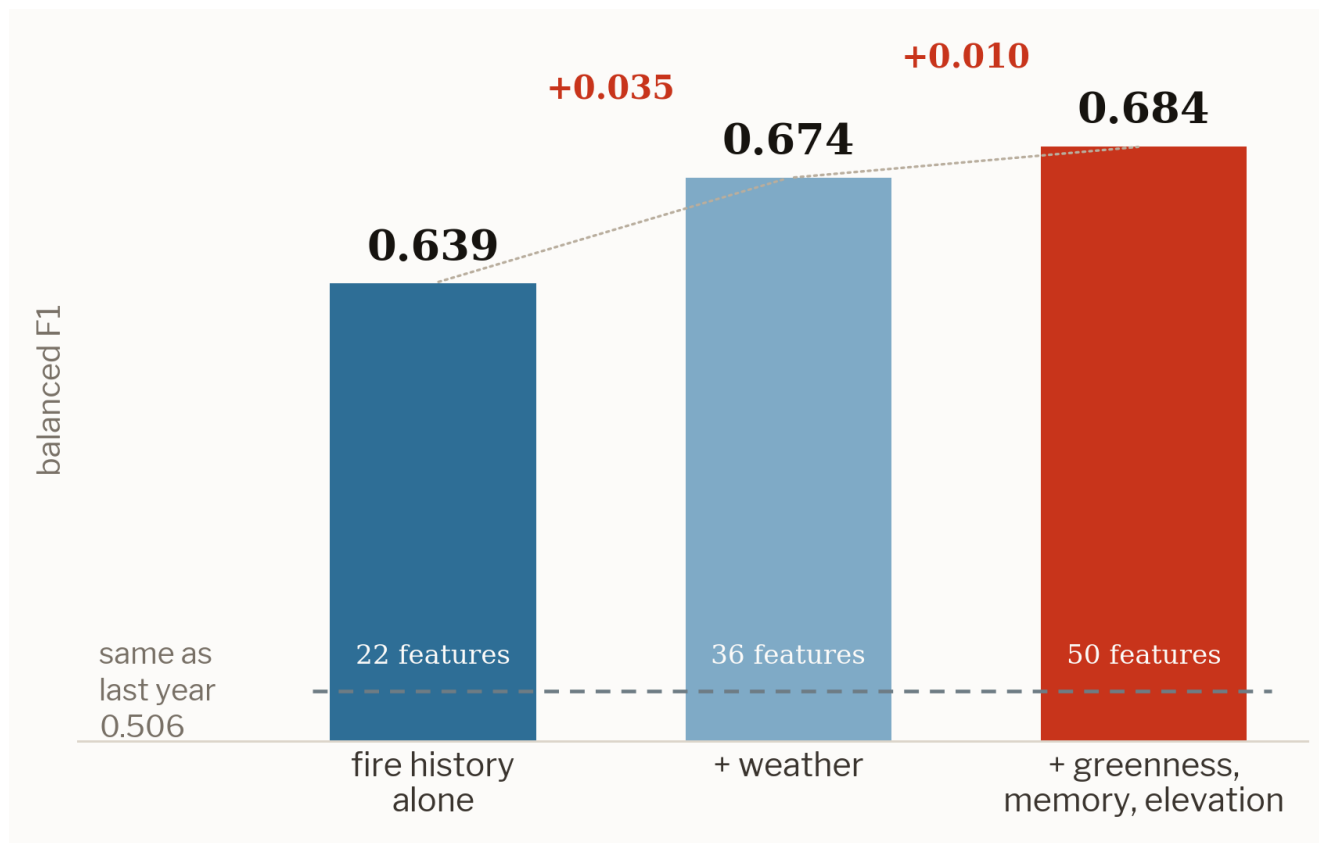
Read it this way

Balanced F1 treats all three answers as equally important, so a model cannot score well by ignoring the rare ones.

The number to watch is the gap to “same as last year” at 0.506. That is what a fire officer already knows without us. Gradient boosting adds 0.178 on top of it.

Accuracy on the same years is 0.725.

Weather earned its place. Greenness earned less, and we said so.



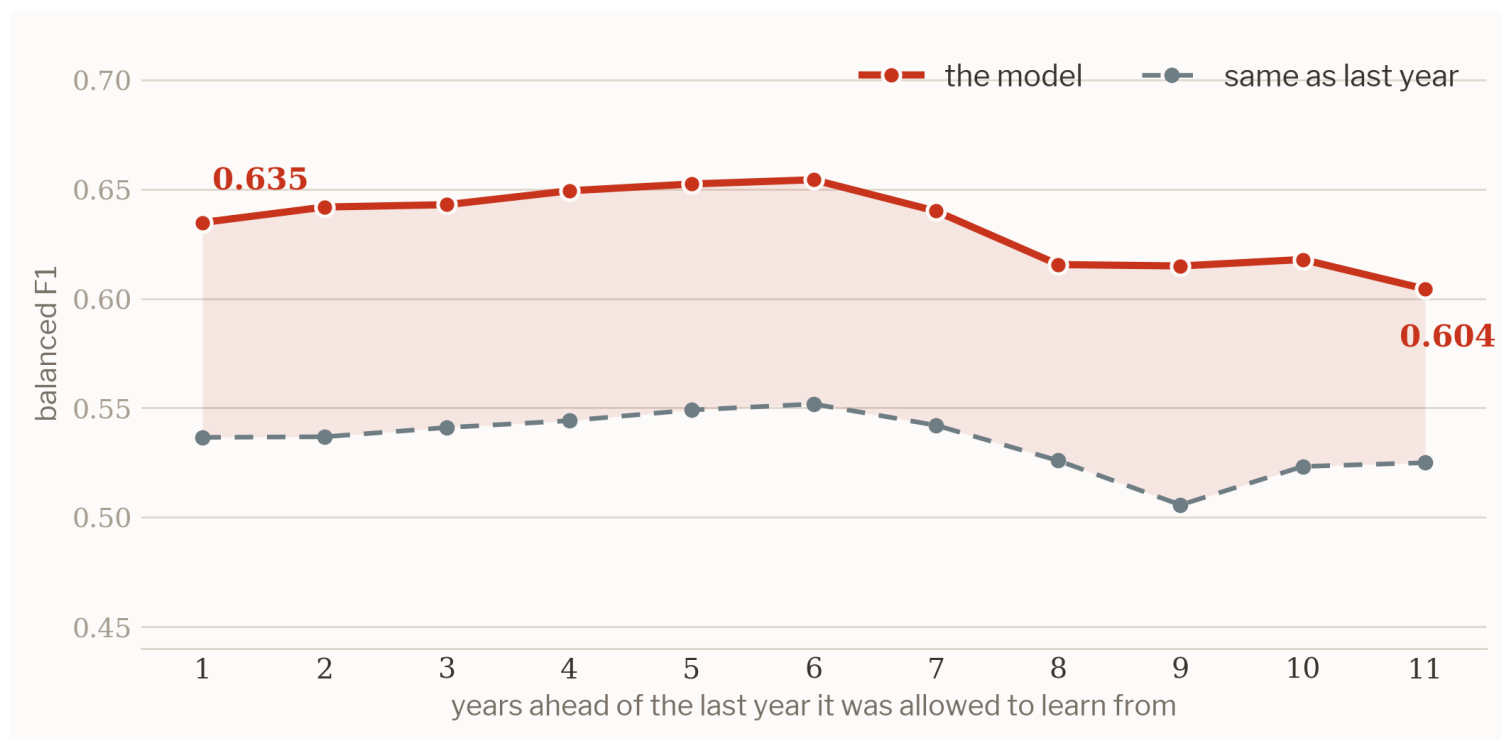
Adding the weather was the single biggest step in the project — +0.035, from knowing how wet the wet season was.

The third run adds greenness, longer fire memory and elevation together, for +0.010. Isolated in the experiment ledger, greenness on its own is worth +0.029.

We kept it, because it holds up on years the model never saw and because two of the model's five strongest inputs turn out to be greenness — even though Earth Engine cost ten minutes a run against sixty seconds for everything else.

One good year could be luck. So we did it sixty-six times.

We cut the record at each year from 2014 to 2024, let the model learn only what came before the cut, and scored it on every year after — 66 separate tests.



The surprise

Skill does not fade with distance. Guessing one year ahead scores 0.635; eleven years ahead it still scores 0.604.

That is not the model being clever. It is telling us the strongest clue is the wet season just gone, which is known by the end of April however old the fire history is.

We tuned every model, and checked it would hold up.

Four checks, all scored on years the models had not seen.

Tuned properly

0.684 → 0.684

A search over every model's settings, scored on the validation years only, left gradient boosting where it was. The decision tree gained most: 0.604 → 0.632.

Checked for overfitting

0.821 → 0.684

Gradient boosting on the years it trained on, then on years it never saw. The random forest memorises far more: 0.924 → 0.660.

Scored beyond accuracy

ROC-AUC 0.862 · PR-AUC 0.732

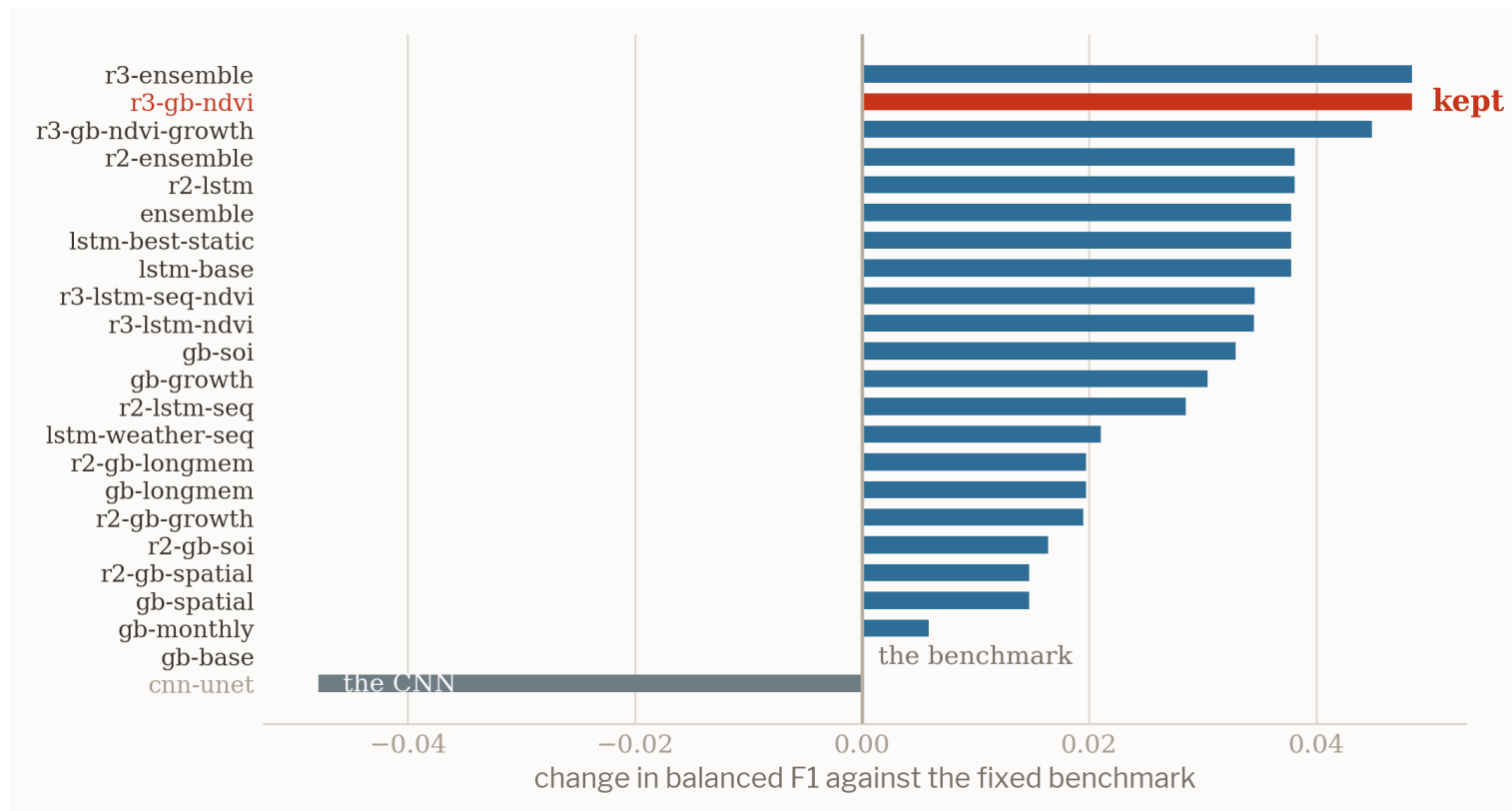
These score the probabilities behind each answer. On late fires the PR-AUC is 0.558, about twice what guessing would get.

Why not shuffled cross-validation

0.772 vs 0.677

Shuffled 5-fold cross-validation leaks between neighbouring years and overstates the score. Testing forward in time gives the real figure.

Twenty-three experiments against a fixed benchmark. One was kept.



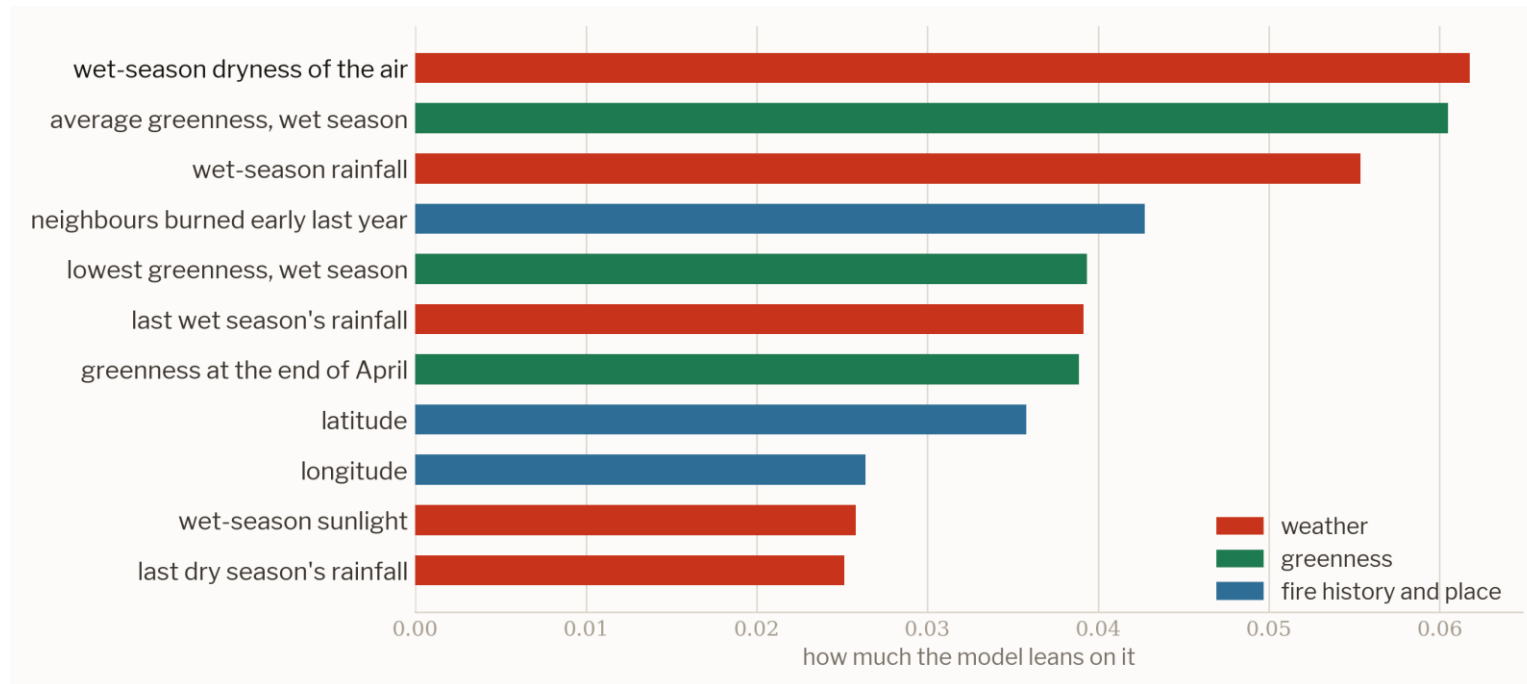
Including the one that failed

We built a U-Net over the whole grid to let a convolutional network read fire as a picture rather than a table of numbers.

It scored 0.588 — 0.048 below the benchmark — so it went in the ledger as a reverted run and stayed out of the final model.

Every run was scored the same way, against the same fixed benchmark, and recorded whether it helped or not.

The model rediscovered what land managers already knew.



Nobody told it that a wet, hot wet season grows the grass that carries an October fire. It found that in the data.

The top five inputs are the dryness of the air, average greenness, wet-season rainfall, what the neighbouring country did last year, and the lowest greenness of the wet.

That it lands where fire ecology already sits is the best evidence we have that it learned the problem rather than the dataset.

What this does not do.

Late fires are still the hardest call

balanced F1 0.553

The class that matters most is the one we predict worst. It is the rarest and the most weather-driven, and a year's weather after April is exactly what we refuse to look at.

A season, not a day

This says which squares are likely to burn and roughly when. It does not say which Tuesday, and it is not a fire-danger rating.

2026 is unverified

No burnt-area record exists for a year still running. The forecast is published as a claim that can be checked in 2027, not as a result.

One Territory, one grid

Everything is fitted to the NT at 0.05°. Whether it transfers to the Kimberley or Cape York is an open question, not an assumption.

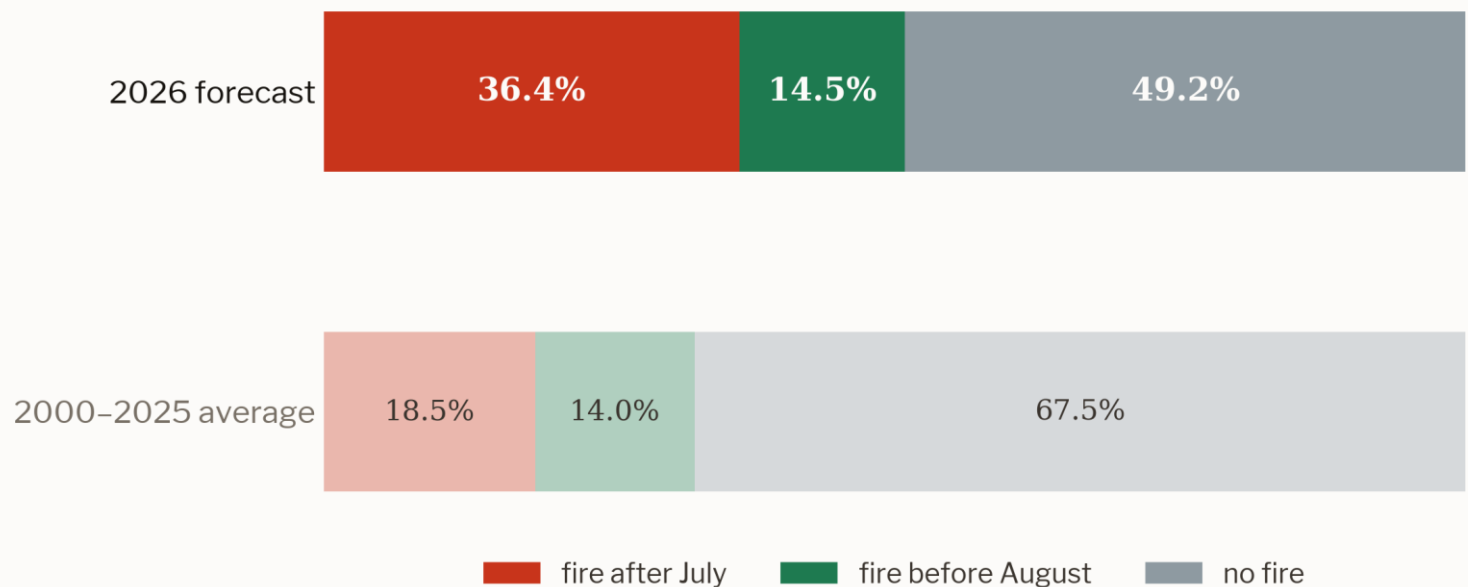
ACT FOUR

What it says about 2026

A forecast that can be checked next year, and the product that puts it in front of someone who is not a data scientist.

Harsh Rastogi

2026 is running heavy, and most of what is left will burn late.



51%

of the Territory forecast to burn in 2026, against a 26-year average of 33%

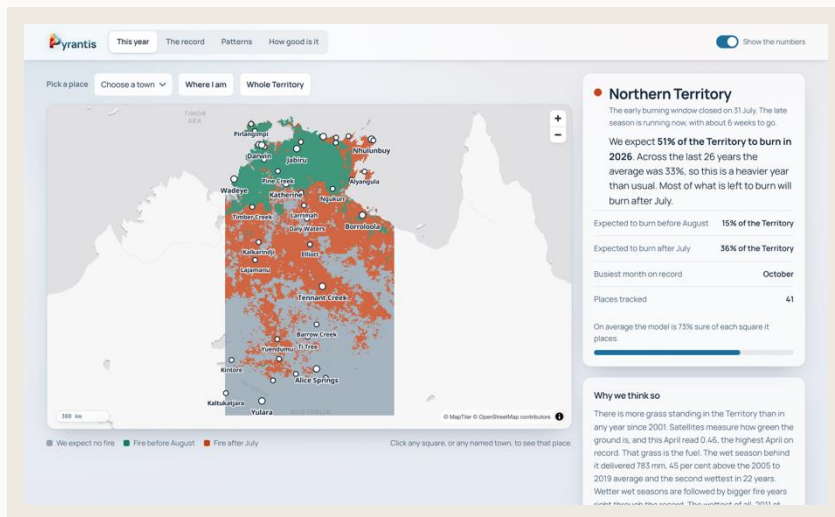
The model reads a wet season that delivered 783 mm — 45% above the 2005 to 2019 average and the second wettest in 22 years — and the greenest April on record at 0.46.

More water grew more grass. More grass is more fuel.

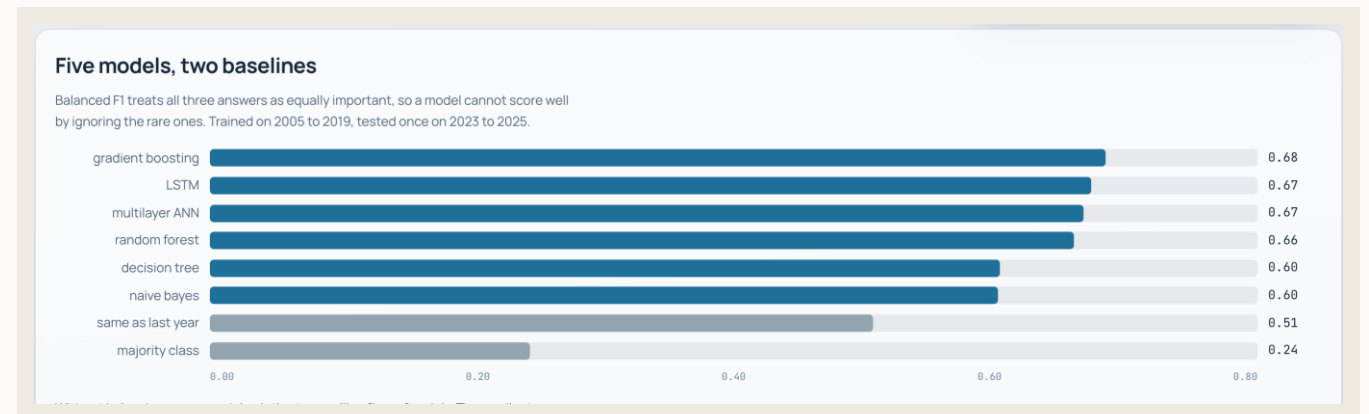
Mean confidence across the map is 0.733. No fire scars exist for 2026 yet, so this is a claim we can be held to in 2027.

A model in a notebook helps nobody.

So we shipped it: a public site that answers the question in plain English first, and shows the map, the record and the evidence underneath.



This year — the forecast for any town or square



How good is it — every model, scored on years it never saw

Every square, every year, and an honest number on how often it is right.

Fire in the Territory is not going to stop. Knowing in April which country is likely to carry an October fire is how a burning programme gets planned while there is still time to plan it.

pyrantis.harshlabs.workers.dev

github.com/harshrastogii/PYRANTIS

Every figure in this deck is generated from the repository, not typed in.

0.684

balanced F1, years never seen

66

independent re-tests

51%

forecast to burn in 2026

References (1 of 2)

American Psychological Association, 7th edition

- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>
- Chollet, F., & others. (2015). Keras [Computer software]. <https://keras.io>
- Didan, K. (2021). MODIS/Terra vegetation indices 16-day L3 global 250 m SIN grid V061 [Data set]. NASA EOSDIS Land Processes Distributed Active Archive Center. <https://doi.org/10.5067/MODIS/MOD13Q1.061>
- Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29(5), 1189–1232. <https://doi.org/10.1214/aos/1013203451>
- Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., & Moore, R. (2017). Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment*, 202, 18–27. <https://doi.org/10.1016/j.rse.2017.06.031>
- Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., ... Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, 585(7825), 357–362. <https://doi.org/10.1038/s41586-020-2649-2>
- Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>
- Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3), 90–95. <https://doi.org/10.1109/MCSE.2007.55>

References (2 of 2)

American Psychological Association, 7th edition

- Jeffrey, S. J., Carter, J. O., Moodie, K. B., & Beswick, A. R. (2001). Using spatial interpolation to construct a comprehensive archive of Australian climate data. *Environmental Modelling & Software*, 16(4), 309–330. [https://doi.org/10.1016/S1364-8152\(01\)00008-1](https://doi.org/10.1016/S1364-8152(01)00008-1)
- North Australian Fire Information. (2026). Northern Territory fire scar mapping [Data set]. Darwin Centre for Bushfire Research, Charles Darwin University. <https://firenorth.org.au>
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- Queensland Government. (2026). SILO climate data: Data Drill [Data set]. Long Paddock. <https://www.longpaddock.qld.gov.au/silo/>
- Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. In N. Navab, J. Hornegger, W. M. Wells, & A. F. Frangi (Eds.), *Medical image computing and computer-assisted intervention – MICCAI 2015* (pp. 234–241). Springer. https://doi.org/10.1007/978-3-319-24574-4_28
- Russell-Smith, J., Yates, C. P., Whitehead, P. J., Smith, R., Craig, R., Allan, G. E., Thackway, R., Frakes, I., Cridland, S., Meyer, M. C. P., & Gill, A. M. (2007). Bushfires 'down under': Patterns and implications of contemporary Australian landscape burning. *International Journal of Wildland Fire*, 16(4), 361–377. <https://doi.org/10.1071/WF07018>